## 159102 workshop – Week 1

**1.** In each case, **given a truth table**, find an equivalent Boolean expression, then draw an associated logic circuit.

**a)**

| A | B | output |
|---|---|---|
| F | F | T |
| F | T | F |
| T | F | F |
| T | T | T |

**b)**

| A | B | C |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

**c)**

| A | B | output |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 0 |

**d)** What would you do if the outputs of the truth table were all 1 (or all T)?

What if the outputs of the truth table were all 0 (or all F)?

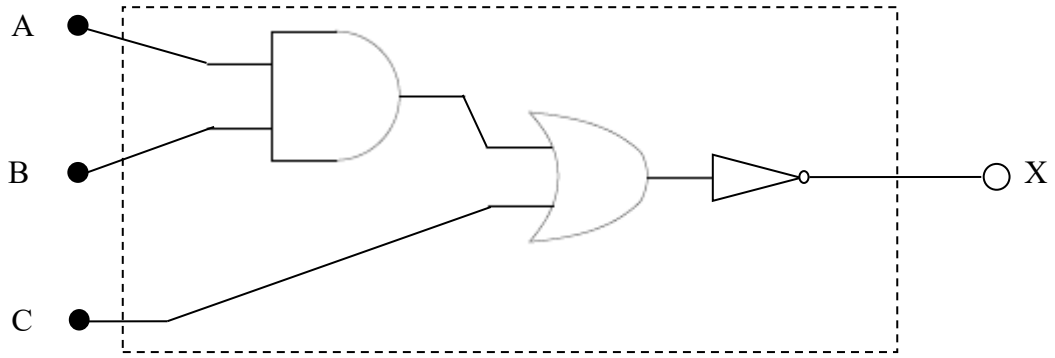
**e)**

| A | B | C | X |
|---|---|---|---|
| F | F | F | F |
| F | F | T | T |
| F | T | F | F |
| F | T | T | T |
| T | F | F | F |
| T | F | T | F |
| T | T | F | F |
| T | T | T | T |

**f)**

| A | B | C | D |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | 1 |
| 1 | 1 | 1 | 0 |

**2.** Given the logic circuit below, find the associated truth table and Boolean expression:




**3.** Given the following Boolean expression, draw the circuit and set up the truth table:

X = ~A ● B ● C + ~A ● B ● ~C + A ● ~B ● C

**4.** Given D = ~(A ● B ● C + A) + A ● B + A ● ~C

a) find the value of D if A is true, B is false, and C is false

b) find the value of D if A, B and C are all true